ARNAV SABHARWAL
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EDUCATION

Carnegie Mellon University School of Computer Science Pittsburgh, PA
MSc in Computer Science May 2027

BSc in Computer Science, Concentration in Principles of Programming Languages. University Honors and School of Computer Science College Honors. May 2026

RESEARCH EXPERIENCE

Fifth Year M.S. Research, Advisor: Jan Hoffmann May 2026 – May 2027
Enabling analysis of memoized functions in Resource Aware ML 2, a tool for statically bounding the cost of ML programs

Summer Research + Undergraduate Honors Research Thesis, Advisor: Prof. Jan Hoffmann May 2025 – May 2026
Implementing hybrid resource analysis in Resource Aware ML 2

Independent Study, Advisor: Prof. Jeremy Avigad December 2023 – May 2024
Prototyped a solution to the absence of co-inductive types (e.g., infinite streams) in the Lean 4 theorem prover

PUBLICATIONS

Implementing Hybrid Resource Analysis in Resource Aware ML 2 June 2026
Presented at the PLDI 2026 Student Research Competition, won third place in undergraduate category

PROJECTS

Resource-aware Programming Languages (15-714) Fall 2025
Mechanized an important property of LFPL, a language that only admits polynomial time computations, in Lean 4

Compiler Design (15-411) Spring 2025
Wrote an optimizing compiler targeting X86-64 for C0, a safe subset of C, in Rust

TEACHING ASSISTANTSHIPS

Each TA role below involves designing and grading homework and exams, responding to student questions in the class forum and during weekly in-person office hours, and designing and teaching recitations and exam reviews.

Foundations of Programming Languages (15-312) Fall 2025
Parallel and Sequential Data Structures and Algorithms (15-210) Fall 2024
Principles of Imperative Computation (15-122) Summer 2023, Fall 2023

WORK EXPERIENCE

Ernst and Young LLC Mumbai, India
Software Engineer May 2024 – August 2024
Implemented & evaluated a retrieval augmented generation pipeline with a novel knowledge-graph construction method